

DIABETIC RETINOPATHY CLINICAL DECISION SUPPORT SYSTEM

Ophthalmic Retinal Screening & Decision-Support Consultation Summary

Target Standard: FDA SaMD Guidance · NHS DTAC / NICE ESF Clinical Quality Protocols

Assessment ID:	REC-2026-6885	Patient Identifier:	PT-TEST-001
Eye Laterality:	OD (Right Eye)	Acquisition Date:	2026-09-26 20:29 UTC
Camera Model:	Topcon TRC-NW400	Dilation Protocol:	Non-Mydriatic

1. Technical Image Quality & 3-Stage Validation Pipeline

Gate 1: File Integrity	PASSED	MIME image/jpeg, SHA-256: 4b3671e87b916fbb...
Gate 2: Retinal Relevance	PASSED	Retinal aperture geometry and spectral balance confirmed.
Gate 3: Technical Quality	PASSED	Laplacian variance: 670.7 (Threshold >= 60.0), Illum: 1.00

2. Preliminary Model-Generated Observation (AI Assistive Engine)

Primary Candidate Class:	Moderate NPDR	Class Relative Score:	0.78
Model Identity:	EfficientNet-B0 (Frozen Evaluation)	Target Saliency Layer:	features.8 (Conv2d Bottleneck Residual)

ICDR Stage	Grade Description	Model-Generated Class Score
Grade 0	Grade 0: No Apparent DR	0.04
Grade 1	Grade 1: Mild NPDR	0.12
Grade 2	Grade 2: Moderate NPDR	0.78
Grade 3	Grade 3: Severe NPDR	0.05
Grade 4	Grade 4: Proliferative DR	0.01

Regulatory Boundary Notice: Scores reflect feature activations derived from the pre-trained EfficientNet-B0 network. Scores do not represent clinical certainty, diagnostic probability, or confirmed diagnosis.

3. Certified Clinical Evaluation (Authoritative Human-in-the-Loop)

Reviewing Clinician:	Dr. Adaeze Okonjo, MBChB, FRCOphth	License / GMC Number:	GMC-7492104
Clinical Agreement:	AGREED with Model Observation	Certified ICDR Grade:	Grade 2: Moderate NPDR
Clinical Notes / Rationale:	Confirmed presence of parafoveal microaneurysms.	Management / Referral:	Routine recall in 6 months.
Signature Timestamp:	2026-09-26 20:29:54 UTC	Cryptographic Hash:	SIG-SHA256-ab318b8aead5ec90c034119f

CONFIDENTIAL CLINICAL CONSULTATION RECORD: This document is generated for clinical decision support. The certified diagnosis and management decisions are solely the legal and medical responsibility of the named reviewing practitioner. Original photographic evidence cryptographically anchored via SHA-256 digest.